

1(b)(iii)	1 horizontal line at a non-zero value of a .	B1
	2 curved line from origin with increasing gradient.	B1
1(c)	acceleration (of free fall) is unchanged / not dependent on mass <u>and</u> so no effect (on time taken).	A1

Question	Answer	Mark
2(a)(i)	force $\times$ distance <u>moved</u> in the direction of the force	B1
2(a)(ii)	energy (of a mass/body) due to motion / speed / velocity	B1
2(b)(i)	1 $E = \frac{1}{2}mv^2$	C1
	$(\Delta)E = \frac{1}{2} \times 580 \times (22^2 - 12^2) = 9.9 \times 10^4 \text{ J}$	A1
	2 $(\Delta)E = mg(\Delta)h$ $\Delta E = 580 \times 9.81 \times 13$	C1
	$= 7.4 \times 10^4 \text{ J}$	A1

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Question	Answer	Marks
2(b)(ii)	length = $(2\pi \times 13) / 4$ or $(\pi \times 26) / 4$ or $(\pi \times 13) / 2 = 20$ m	A1
2(b)(iii)	work done against resistive force = $9.9 \times 10^4 - 7.4 \times 10^4$ average resistive force = $(9.9 \times 10^4 - 7.4 \times 10^4) / 20$	C1
	= 1300 N	A1
2(b)(iv)	from horizontal/right to vertical / up or 90°	A1
2(b)(v)	$p = mv$ or (580×22) or (580×12)	C1
	$\Delta p = [(580 \times 12)^2 + (580 \times 22)^2]^{0.5}$	C1
	= 1.5×10^4 N s	A1

Question	Answer	Marks
3(a)(i)	force / (cross-sectional) area	B1